THE IMAGE OF A CLOSED NOWHERE DENSE SUBSET UNDER A FINITE MORPHISM IS AGAIN CLOSED AND NOWHERE DENSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-image-nowhere-dense-finite is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 12499-12504 and proof span 12506-12570 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains an xymatrix diagram with five arrows; it is copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. The proof's closing phrase 'and we win' is the author's and is preserved.

Editorial note. 8 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A finite morphism is proper, hence closed, and the inverse image of a closed nowhere dense subset under an open morphism is nowhere dense. Cited at proof lines 12506-12508 and 12524-12529. Gives closedness of the image and lets the nowhere density be transported along the open morphisms appearing in the auxiliary diagram.
- Nowhere density is local on the target, and a finite union of closed nowhere dense subsets is closed and nowhere dense. Cited at proof lines 12511-12513 and 12545-12548. Reduces the statement to an affine base and assembles the final case from finitely many closed pieces.
- An open inverse image of a closed subset is closed, and the image of a closed nowhere dense subset under a closed immersion stays closed and nowhere dense. Cited at proof lines 12530-12533. Allows the auxiliary diagram to be replaced by the situation where the source is a finite union of closed pieces mapping isomorphically to the base.
- The auxiliary diagram for a finite morphism over an affine base, with a finite locally free morphism to the target and to the source, cited from

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the same chapter. Cited at proof lines 12516-12518. Produces the open morphisms along which the problem is invariant.
- A finite locally free morphism is flat, and a flat morphism locally of finite presentation is open, cited from the same chapter. Cited at proof lines 12521-12524. Shows that the morphisms of the auxiliary diagram are open.

Source reference. Original lemma 02KB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3366–3366. Same-chapter reference used in the retained author text at source line(s) 12527, 13544. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01WN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2488–2488. Same-chapter reference used in the retained author text at source line(s) 11282, 12507, 13352. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2403–2403. Same-chapter reference used in the retained author text at source line(s) 4492, 4621, 5693, 5890, 7179, 8281, 12527. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03HW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4582–4582. Same-chapter reference used in the retained author text at source line(s) 12525. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03HQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4576–4576. Same-chapter reference used in the retained author text at source line(s) 12535. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03HO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4574–4574. Same-chapter reference used in the retained author text at source line(s) 12547, 15267. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03HP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4575–4575. Same-chapter reference used in the retained author text at source line(s) 12513. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03HR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4577–4577. Same-chapter reference used in the retained author text at source line(s) 12529, 12533. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : Y \to X$ be a finite morphism of schemes. Let $T \subset Y$ be a closed nowhere dense subset of Y. Then $f(T) \subset X$ is a closed nowhere dense subset of X.*

Proof. By Lemma 01WN we know that $f(T) \subset X$ is closed. Let $X = \bigcup X_i$ be an affine covering. Since T is nowhere dense in Y, we see that also $T \cap f^{-1}(X_i)$ is nowhere dense in $f^{-1}(X_i)$. Hence if we can prove the theorem in the affine case, then we see that $f(T) \cap X_i$ is nowhere dense. This then implies that T is nowhere dense in X by Topology, Lemma 03HP.

Assume X is affine. Choose a diagram

$$\begin{array}{ccccc} Z' & \xleftarrow{i} & Y' & \xrightarrow{a} & Y \\ & \searrow & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ & & X' & \xrightarrow{b} & X \end{array}$$

as in Lemma 03HW. The morphisms a, b are open since they are finite locally free (Lemmas 02KB and 01UA). Hence $T' = a^{-1}(T)$ is nowhere dense, see Topology, Lemma 03HR. The morphism b is surjective and open. Hence, if we can prove $f'(T') = b^{-1}(f(T))$ is nowhere dense, then $f(T)$ is nowhere dense, see Topology, Lemma 03HR. As i is a closed immersion, by Topology, Lemma 03HQ we see that $i(T') \subset Z'$ is closed and nowhere dense. Thus we have reduced the problem to the case discussed in the following paragraph.

Assume that $Y = \bigcup_{i=1,\ldots,n} Y_i$ is a finite union of closed subsets, each mapping isomorphically to X. Consider $T_i = Y_i \cap T$. If each of the T_i is nowhere dense in Y_i, then each $f(T_i)$ is nowhere dense in X as $Y_i \to X$ is an isomorphism. Hence $f(T) = f(T_i)$ is a finite union of nowhere dense closed subsets of X and we win, see Topology, Lemma 03HO. Suppose not, say T_1 contains a nonempty open $V \subset Y_1$. We are going to show this leads to a contradiction. Consider $Y_2 \cap V \subset V$. This is either a proper closed subset, or equal to V. In the first case we replace V by $V \setminus V \cap Y_2$, so $V \subset T_1$ is open in Y_1 and does not meet Y_2. In the second case we have $V \subset Y_1 \cap Y_2$ is open in both Y_1 and Y_2. Repeat sequentially with $i = 3, \ldots, n$. The result is a disjoint union decomposition

$$\{1, \ldots, n\} = I_1 \amalg I_2, \quad 1 \in I_1$$

and an open V of Y_1 contained in T_1 such that $V \subset Y_i$ for $i \in I_1$ and $V \cap Y_i = \emptyset$ for $i \in I_2$. Set $U = f(V)$. This is an open of X since $f|_{Y_1} : Y_1 \to X$ is an isomorphism. Then

$$f^{-1}(U) = V \amalg \bigcup\nolimits_{i \in I_2} (Y_i \cap f^{-1}(U))$$

As $\bigcup_{i \in I_2} Y_i$ is closed, this implies that $V \subset f^{-1}(U)$ is open, hence $V \subset Y$ is open. This contradicts the assumption that T is nowhere dense in Y, as desired. $\square$